

How surprising are Austro-Tai lookalikes? Meaning-blind permutation tests of Smith’s (2025) reconstructions and of dual-attested Lexibank form inventories

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Companion code and data: github.com/jeremycollinsmpi/austro-tai-llm-test.

Abstract

The Austro-Tai hypothesis posits a genetic link between Austronesian and Kra-Dai (Tai-Kadai). Published lexical comparisons are hard to quantify: lists may mix robust etymologies with weakly justified or lookalike-friendly reconstructions, and similarity judgments inflate when meanings are known. We report two meaning-blind permutation screens in which an LLM (gpt-4.1, via API) scores segmental form similarity without seeing meanings. **Study 1** audits Alexander D. Smith’s (2025) gloss-aligned Proto-Austronesian (PAN) and Proto-Kra-Dai (PKD) reconstructions: after Lexibank attestation filters, 27 of 79 **Tier A** pairs—those with adequate Lexibank coverage and attestation scores ≥ 2 for both PAN and PKD—meet a generous hit threshold against a null mean of 5.7 (100 permutations; one-sided $p \approx 0.01$). **Study 2** is designed to be more robust to reconstruction cherry-picking. It compares phylogenetically sampled modern Lexibank form groups for the same Concepticon meaning—restricted to dual-attested concepts on the Blust/ABVD basic-vocabulary list (194 concepts)—and shuffles Austronesian form groups across Tai-Kadai slots. Hits at generosity $\geq 4 / \geq 3 / \geq 2$ are 11 / 17 / 121 against null means of about 2.2 / 5.7 / 91.7 (30 permutations; $p \approx 0.032$ at each of those thresholds). Convergence across a reconstruction audit and a reconstruction-free Lexibank screen supports excess form resemblance under these controls—not genetic proof, which requires systematic sound correspondences. We situate the results in the Austro-Tai literature (Benedict; Ostapirat; Sagart) and discuss limits of LLM-based screening.

Keywords: Austro-Tai; Kra-Dai; Austronesian; lexical comparison; chance resemblance; permutation test; Lexibank; large language models; Blust basic vocabulary

1. Introduction

Macro-comparative hypotheses live or die on lexical and phonological evidence, yet it remains surprisingly hard to say **how strong** a published lookalike package is. Impressive lists can mix secure etymologies with weakly justified reconstructions; judgments of “similarity” inflate when meanings are visible; and without an explicit chance baseline, it is unclear whether the published pairing of forms across two proto-languages exceeds what generous form matching would produce under random reassignment.

The **Austro-Tai** hypothesis—positing a genetic link between Austronesian and Kra-Dai (Tai-Kadai)—is a long-standing case in point. From Benedict’s early lexical proposals through Ostapirat’s correspondence-based arguments and Sagart’s phylogenetic alternatives, debate has turned on whether shared vocabulary reflects inheritance, contact, or chance (see §2). What has been rarer is a statistical audit under controls that address two confounders at once: (i) **semantic priming** in cognate-style judgments, and (ii) **selection of favorable forms**, whether as weakly attested reconstructions or as cherry-picked lookalikes among modern languages.

This paper does not assemble a new cognate list, propose regular sound correspondences, or adjudicate sister-family vs Austronesian-internal geometries for Kra-Dai. It asks two related questions. **Study 1** takes Alexander D. Smith’s (2025) open Zenodo package of gloss-aligned Proto-Austronesian (PAN) and Proto-Kra-Dai (PKD) reconstructions as the object of evaluation: how much form resemblance remains—and how surprising is it—after weakly attested reconstructions are set

aside? **Study 2** asks a stricter follow-up that does not depend on any reconstruction package: among well-attested Lexibank concepts, do **groups of modern Tai-Kadai and Austronesian forms** for the *same* meaning show more shape resemblance than form groups for *mismatched* meanings under a meaning-blind screen?

Study 1 has two layers. **Layer 1** scores each reconstruction against modern Lexibank forms (all Tai-Kadai daughters for PKD; a phylogenetically stratified Austronesian sample for PAN) and excludes coverage gaps and attestation score 1 from the analysis set (Tier A). **Layer 2** compares PAN and PKD **forms only** (meanings withheld) with a fixed LLM screen and tests whether high similarity scores exceed a null in which PAN shapes are randomly reassigned across PKD slots. On **79** Tier A pairs we observe **27** hits against a null mean of **5.7** (100 permutations; one-sided $p \approx 0.01$).

Study 2 is designed to be more robust to reconstruction cherry-picking. Concept selection follows Lexibank coverage and the Austronesian Basic Vocabulary (Blust/ABVD) Concepticon list rather than an author's comparative spreadsheet; within each concept we draw phylogenetically stratified modern samples (with onomatopoeia and near-duplicate filters) and score **set-versus-set** shape resemblance without showing the gloss. The null shuffles Austronesian form groups across Tai-Kadai slots. On **194** dual-attested Blust concepts, observed hits at generosity ≥ 4 are **11** against a null mean of **2.2** (30 permutations; one-sided $p \approx 0.032$); the same floor p holds at ≥ 3 (17 vs 5.7) and even at the liberal ≥ 2 threshold (121 vs 91.7).

We interpret both results as quantified evidence of **excess form resemblance under stated controls**—not as proof of genetic relationship, which would require systematic sound correspondences. Study 2's convergence with Study 1 is especially informative because it does not inherit Smith's choice of which proto-forms to align.

Section 2 surveys the Austro-Tai literature and justifies using Smith (2025) for Study 1. Section 3 describes the data. Section 4 details methods for both studies. Section 5 reports results. Section 6 discusses interpretation and limits; Section 7 concludes.

2. Background: Austro-Tai and the assessment of lexical evidence

2.1 Literature map

Early proposal and controversy

Paul K. Benedict (1942, 1975) advanced **Austro-Tai** as a deep genetic relationship linking Austronesian with what is now usually called Kra-Dai (Tai-Kadai), drawing heavily on lexical lookalikes across basic and cultural vocabulary. The proposal was influential in framing a research program, and equally influential as a cautionary example in methodological debates: multilateral comparison without tightly controlled sound correspondences invites chance resemblance, semantic stretch, and post hoc selection of favorable pairs. Whatever one's view of Benedict's specific etymologies, the enduring problem he raises for later work is how to tell **how much** of a published lookalike package exceeds what chance and liberal similarity judgments would produce.

Regular correspondences and a sister-family construal

A major modern strand argues that Kra-Dai and Austronesian are related as **sister families** (or close relatives) on the strength of basic vocabulary **and** systematic phonological correspondences. Ostapirat (2005, 2013) is central here: the claim is not merely that many forms look alike when meanings are matched, but that segmental (and related) correspondences can be stated with comparative-method discipline. On this view, Austro-Tai is a classical genetic hypothesis to be won or lost on regularity and reconstructibility, not on raw list length. Our study does **not** re-implement Ostapirat's correspondence system; we return to it in the Discussion as the standard against which

“non-chance form resemblance under a generous screen” remains a weaker, preliminary kind of evidence.

Alternative phylogenetic architecture

A second modern strand, associated especially with Sagart (2004, 2005, 2019), places Kra-Dai in a more **Austronesian-internal** (or AN-adjacent) phylogenetic frame—linking numerals, subgrouping arguments, and models of how Kra-Dai tone systems might relate to Austronesian coda history (see also recent tonogenesis discussion such as Liao & Gehrmann 2025 in *Diachronica*). The empirical overlap with Ostapirat’s comparisons can be substantial even when the **tree geometry** differs: both traditions posit more than accidental lexical connection between the families, but they disagree on where Kra-Dai sits relative to Formosan and Malayo-Polynesian diversification. Our permutation result is intentionally **agnostic** on sister-family vs AN-internal architecture; excess form similarity in a published alignment set is compatible with either genealogy once chance browsing is set aside.

Contact, loans, and areal lookalikes

Kra-Dai languages have long histories of contact with Sinitic and with mainland Southeast Asian neighbors. Chinese loans, calques, and areal diffusion can produce meaning-matched form similarities that are not inheritance from a shared Austro-Tai parent. Published reconstruction packages may flag some Chinese comparisons (Smith’s sheet includes Chinese-related columns and notes); our Layer 2 judge never sees those flags or the glosses, but Layer 1 attestation against Lexibank daughters does not by itself separate inheritance from old loans that have become family-internal. Contact therefore remains a live alternative explanation for residual lookalikes and a limit on interpreting aggregate hit counts as phylogenetic proof.

Competing higher groupings (brief)

Austro-Tai is one among several contested higher-order proposals involving Austronesian (e.g. strands of Austric or Sino-Austronesian argumentation). We mention them only to locate Austro-Tai in a crowded macro-comparative landscape: the present paper audits **one** open Austro-Tai package under explicit statistical controls, not the relative merits of every macro-family hypothesis.

Chance resemblance and the need for null models

Methodological critiques of multilateral comparison (notably Ringe and related work) stress that large vocabularies, phonetic latitude, and semantic flexibility make impressive-looking lists easy to assemble. When the object of evaluation is itself a **published list**, the natural statistical question is comparative: relative to a well-defined null, how often would equally “generous” form matches arise if proto-shapes were randomly re-paired across meanings? Explicit permutation or chance baselines remain uncommon for entire Austro-Tai alignment sets—especially after first removing reconstructions that modern daughters do not support. That is the gap this paper addresses.

Table 1 (schematic). Positions in the Austro-Tai debate (compact).

Strand	Exemplars	Core claim (simplified)	Relevance here
Early lexical macro-comparison	Benedict (1942, 1975)	Deep AN-KD link via lookalikes	Motivates quantifying list strength
Sisters + correspondences	Ostapirat (2005, 2013)	Relatedness via regular sound change	Stronger standard than our screen
AN-linked phylogeny / tonogenesis	Sagart (2004–2019); cf. Liao & Gehrmann (2025)	KD placed in/near AN diversification	Compatible with non-chance lexical signal; different tree
Contact / loans	Sinitic and mainland SEA contact traditions	Lookalikes need not be genetic	Limits causal interpretation of hits
Chance / method critique	Ringe and related	Need nulls and tight controls	Motivates permutation design

2.2 Justification for using Smith (2025)

We take as our object of evaluation Alexander D. Smith’s (2025) *Austro-Tai comparative dataset (Reconstructions)* (Zenodo), rather than compiling a new etymological list. Three properties motivate that choice.

First, the spreadsheet provides **gloss-aligned** Proto-Austronesian and Proto-Kra-Dai reconstructions in a single package explicitly aimed at Austro-Tai comparison. That structure makes a slot-wise permutation test well-defined: PKD forms (and meaning slots) can be held fixed while PAN forms are shuffled across slots, so that the null world preserves the same inventory of shapes and asks only whether the **published pairing** is special.

Second, the release is **open, versioned, and preliminary** by the author’s own framing. Preliminary reconstruction packages invite exactly the stress test we perform: external checks on whether proposed proto-forms are recoverable from modern Lexibank daughters, and a quantified assessment of residual form similarity after weak reconstructions are set aside. Branch-level Kra-Dai columns in the sheet (Proto-Tai, Proto-Hlai, etc.) supply an author-internal signal of comparative breadth; we report them descriptively but do not use them as our primary eligibility filter, so that Tier A is defined by Lexibank-facing criteria rather than by reifying Smith’s branch coding.

Third, anchoring the study in an independent published dataset reduces **researcher degrees of freedom** in selecting which comparisons count as evidence. Had we assembled our own cognate list, critics could fairly ask whether the list was tuned to survive the permutation test. By freezing Smith’s alignments as the object of audit, the researcher’s main choices become the attestation rules, the meaning-blind prompt, and the hit threshold—choices we state explicitly and can vary in robustness checks—rather than which etymologies enter the denominator.

We do **not** treat Smith (2025) as consensus Proto-Kra-Dai, as a definitive PAN lexicon, or as superseding Ostapirat or Sagart. We treat it as a **contemporary evidence package**: a transparent inventory of what has been presented for Austro-Tai comparison in an openly citable form. Our question is not whether every Smith reconstruction is correct, but how much residual PAN-PKD

form similarity remains—and how surprising that residual is under a meaning-blind null—after reconstructions unsupported by Lexibank daughters are removed.

3. Data

3.1 Smith’s (2025) Austro-Tai reconstruction package

Our comparative object is Alexander D. Smith’s openly released *Austro-Tai comparative dataset (Reconstructions)* on Zenodo (DOI 10.5281/zenodo.15597357; we use release **v1.1**). The spreadsheet lists gloss-aligned reconstructions aimed at Austro-Tai comparison. We retain a row when both a Proto-Austronesian (PAN) string and a Proto-Kra-Dai (PKD) string are present, yielding **120** pairs after parsing.

Besides the two proto-forms, the sheet records branch-level Kra-Dai material (e.g. Proto-Tai, Proto-Hlai, Proto-Kam-Sui, and related columns), optional Chinese-loan flags, and free-text notes. We compute a simple **branch attestation count** (how many Kra-Dai branch columns are non-empty) as descriptive metadata. That count is **not** used as the primary gate for Tier A membership: Layer 1 filters (Lexibank coverage and attestation scores) are defined independently of Smith’s branch columns, so that our exclusions do not merely rediscover the author’s own internal breadth coding.

We treat the package as a **published evidence inventory**, not as a consensus reconstruction of Proto-Kra-Dai or as a replacement for Ostapirat’s or Sagart’s comparative proposals. Justification for anchoring the study in this release appears in §2.2.

3.2 Lexibank modern forms

Modern forms used to audit reconstructions come from Lexibank 2, specifically the lexibank-analysed release **v2.2**. Forms are Concepticon-linked. We map each Smith gloss to a Lexibank concept ID with an **explicit** dictionary (no fuzzy string matching), to avoid systematic mis-links (e.g. *blow* → BLOW (OF WIND) rather than LOW). Unmapped glosses are coverage gaps.

- **Tai-Kadai side.** For PKD attestation we use **all** Lexibank forms whose language is classified as Tai-Kadai for the mapped concept.
- **Austronesian side.** Lexibank includes on the order of **978** Austronesian languages. Using every form for every concept would be costly and would overweight Oceanic. We therefore assign each Lexibank Austronesian language a coarse Glottolog-derived clade (Formosan; Western Malayo-Polynesian; South Halmahera–West New Guinea; Oceanic) and draw a **phylogenetically stratified** sample of up to **80** languages per concept (floors 12 / 20 / 8 / 20 across those clades where available; at most one form per language; remainder filled by round-robin). Details of sampling and attestation prompts are in §4.

3.3 Analysis inventories

From the 120 Smith pairs we derive:

Inventory	Role
Coverage-gap pairs	Unmapped concept, or < 3 Lexibank Tai-Kadai or < 3 Austronesian languages for the concept
Unjustified PKDs	PKD attestation_score = 1 vs Lexibank Tai-Kadai forms
Unjustified PANs	PAN attestation_score = 1 vs the stratified Austronesian sample
Tier A	No coverage gap; PKD and PAN scores ≥ 2 — input to meaning-blind judging and permutation

Counts and the path from 120 pairs to **79** Tier A pairs are reported in §5. Software, caches, and machine-readable tables accompanying this paper reproduce these inventories from the same inputs.

3.4 Study 2: Dual-attested Lexibank concepts (no reconstructions)

Study 2 uses Lexibank forms directly. We retain Concepticon concepts with at least **15** Tai-Kadai and **15** Austronesian languages in Lexibank (**373** concepts), then intersect with the Blust (2008) / ABVD 210-item Concepticon list (**194** concepts). Within each concept we draw stratified Tai-Kadai and Austronesian samples (§4.6). Machine-readable inventories: `data/attested_pilot/core_concepts_blust.tsv`, `data/attested_pilot/Blust-2008-210.tsv`.

4. Methods

Our analysis has two studies that share a meaning-blind LLM screen and a permutation null, but differ in what is being paired.

Study 1 (Smith reconstructions) has two layers. **Layer 1** asks whether each published proto-form is supported by modern Lexibank daughters for the same Concepticon meaning. **Layer 2** asks whether, among reconstructions that pass Layer 1, the published PAN–PKD pairings show more **segmental form similarity** than expected if PAN shapes were randomly reassigned across PKD slots.

Study 2 (Lexibank attested forms) drops reconstructions. For dual-attested Concepticon concepts on a Blust/ABVD basic-vocabulary filter, it compares **groups of modern Tai-Kadai and Austronesian forms** for the same meaning against a null that shuffles Austronesian form groups across Tai-Kadai slots. The motivation is robustness: Study 2 cannot inherit cherry-picking of lookalike-friendly proto-forms from a reconstruction spreadsheet.

Both studies withhold meanings from the similarity judge. All LLM calls used the chat endpoint configured for this project with model `gpt-4.1`. Prompts and scoring rules are fixed in the accompanying software release; responses were parsed as JSON, with light sanitization and up to three retries on malformed output. Scores were cached so that identical inputs were never re-queried inconsistently between observed and null analyses.

4.1 Data: Smith alignments and Lexibank

We parsed Alexander D. Smith’s (2025) Austro-Tai reconstruction spreadsheet (Zenodo release v1.1) into gloss-aligned pairs in which both a Proto-Austronesian (PAN) and a Proto-Kra-Dai (PKD) reconstruction are present. After excluding rows without both sides, **120** pairs remained. Smith’s sheet also records branch-level Kra-Dai reflexes (e.g. Proto-Tai, Proto-Hlai, Proto-Kam-Sui); we

retain branch attestation counts as descriptive metadata but do not use them as the primary filter for Tier A membership.

Modern forms come from Lexibank 2 (lexibank-analysed v2.2). Each Smith gloss was mapped to a Lexibank concept ID by an **explicit** dictionary (no fuzzy string matching), to avoid errors such as mapping *blow* to LOW rather than BLOW (OF WIND). For Proto-Kra-Dai validation we used all Lexibank forms whose language is classified as Tai-Kadai. For Proto-Austronesian validation, Lexibank contains on the order of **978** Austronesian languages; sending every form is impractical and would overweight Oceanic. We therefore built a Glottolog-derived coarse clade label for each Lexibank Austronesian language (Formosan; Western Malayo-Polynesian; South Halmahera–West New Guinea; Oceanic) and drew a **phylogenetically stratified sample** of up to **80** languages per concept, with floors of 12 Formosan, 20 Western MP, 8 SHWNG, and 20 Oceanic where available (at most one form per language), filling any remainder by round-robin across clades.

4.2 Layer 1: Reconstruction attestation

For each aligned pair with at least one Lexibank daughter form on the relevant side, we asked the LLM to score how well the proposed proto-form is supported by the attested modern forms for that meaning (*attestation_score* on a 1–5 scale), returning brief notes on supporting and problematic reflexes.

- **PKD attestation** used the full Tai-Kadai form list for the mapped concept (uncapped in this release; typically well under 100 forms).
- **PAN attestation** used the stratified Austronesian sample described above.

We defined a **coverage gap** when the gloss could not be mapped to a Lexibank concept, or when fewer than **three** Tai-Kadai languages or fewer than **three** Austronesian languages attested the concept in Lexibank. Coverage-gap pairs were excluded from the permutation test (Tier A) because the reconstruction cannot be checked against a minimally diverse modern sample.

Separately, we labeled a reconstruction **unjustified** when its attestation score equaled **1** (daughters cluster on shapes incompatible with the proposed proto-form under the model’s assessment). Unjustified PKDs and unjustified PANs remain in the descriptive inventory of Smith’s package but are excluded from Tier A: a meaning-blind “hit” against an unsupported reconstruction would not constitute evidence that the *published comparative package* is well founded.

Tier A—the analysis set for Layer 2—therefore requires: no coverage gap, PKD *attestation_score* ≥ 2 , and PAN *attestation_score* ≥ 2 (when a score is available).

4.3 Layer 2: Meaning-blind form similarity

For each Tier A pair we submitted the PKD form and the PAN form to the LLM **without** the gloss, without Smith’s notes or loanword flags, and without opaque IDs that embed the meaning (batch indices such as 001, 002 were used instead). The system prompt instructed the model to score segmental shape similarity only—as if the PAN string were a candidate Austronesian reconstruction for the same slot as the PKD string—and not to infer or mention meaning. The model returned a **generosity** score from 1 to 5 together with optional notes on plausible segmental correspondences. We count a **hit** when generosity ≥ 4 . (Scores of 3 are treated as non-hits in the primary analysis; a lower threshold can be examined as a robustness check.)

Judgments were cached by the pair (PKD, PAN) under a fixed prompt version. Consequently, the same form combination receives the same score in the observed alignment and in every permutation that recreates it.

4.4 Permutation null

Holding Tier A PKD forms and slot structure fixed, we randomly permuted the Tier A PAN forms across slots ($N = 100$ permutations; fixed random seed). For each permuted world we recomputed the number of hits under the same generosity threshold, reusing the cache whenever a (PKD, PAN) combination had already been scored. The one-sided p -value is

$$[p = ,]$$

where k is the number of permutations whose hit count is at least as large as the observed hit count (add-one smoothing). This null asks how often chance reassignment of PAN shapes among the same PKD slots would produce as many high form-similarity scores as the published pairing. It does **not** test genetic relatedness, regularity of sound change, or the correctness of every reconstruction.

4.5 Scope and non-claims (Study 1)

We do not propose new etymologies, estimate divergence dates, or claim that high generosity scores equal demonstrated cognacy in the comparative-method sense. The LLM is used as a **reproducible generous screen** for segmental resemblance. Systematic sound correspondences—central to classical arguments for Austro-Tai (e.g. Ostapirat 2005)—are not enforced here. Study 1's contribution is a quantified answer to a narrower question: after setting aside reconstructions that Lexibank daughters do not support, how surprising is the residual form similarity in Smith's published alignments under a meaning-blind null?

4.6 Study 2: Dual-attested Lexibank set-versus-set screen

Study 1 still begins from a **published reconstruction package**. Even after Layer 1 filters, critics may worry that Smith's spreadsheet preferentially records lookalike-friendly proto-forms. Study 2 removes reconstructions from the design entirely.

Concept inventory

We considered Lexibank Concepticon concepts attested in at least **15** Tai-Kadai and **15** Austronesian languages (373 concepts). To focus on meanings standardly used in Austronesian comparative work, we intersected this pool with the **Blust (2008) / ABVD 210**-item Concepticon list, yielding **194** dual-attested concepts (exact Concepticon ID match). Sixteen Blust items fall outside the dual-attested ≥ 15 pool, often because Lexibank splits broader Blust glosses (e.g. FOOT OR LEG vs separate FOOT / LEG).

Sampling within concepts

For each concept we built phylogenetically stratified form samples:

- **Tai-Kadai**. Glottolog-derived coarse clades (Tai, Kam-Sui, Kra, Hlai, Lakkia-Biao, Be, other); floors then round-robin fill toward a target of ~40 languages (at most one form per language).
- **Austronesian**. Same clade scheme and floors as Study 1's PAN sample (target ~80).

Before sampling we applied light filters: drop likely onomatopoeic / expressive shapes (very short forms; clear reduplication) and deduplicate near-identical normalized spellings within clade. Tone digits and common separators were ignored in normalization.

Meaning-blind set scoring

The LLM received two **groups** of forms (clade label + form string only)—no gloss, no language names beyond clade tags—and scored hypothetical cognacy from segmental shape alone (generosity 1–5), with instructions to prefer *widespread* shared skeletons over isolated lookalikes and to ignore recognized word meanings. Scores were cached by the group contents under a fixed prompt version.

We report hits at generosity ≥ 2 , ≥ 3 , and ≥ 4 ; the primary interpretive threshold remains ≥ 4 , with ≥ 3 as a secondary cut and ≥ 2 as a liberal sensitivity check.

Permutation null

Holding Tai-Kadai form groups fixed, we randomly reassigned Austronesian form groups across concept slots ($N = 30$; fixed seed). Incomplete permutations were resumable from disk; every null judgment (score, notes, reasoning, and the source concept of the Austronesian form group) was stored so thresholds can be recomputed post hoc. The one-sided p -value uses the same $(k+1)/(N+1)$ estimator as Study 1.

Study 2 therefore tests whether **same-meaning** modern form inventories look more alike, under a generous meaning-blind screen, than **cross-meaning** inventories drawn from the same concept set—without relying on which reconstructions an author chose to publish.

5. Results

5.1 From Smith’s sheet to Tier A

Starting from 120 gloss-aligned PAN–PKD pairs, Layer 1 exclusions left 79 Tier A pairs for the permutation test (Table 1).

Table 1. Sample construction.

Stage	n
Smith pairs with both PAN and PKD	120
Coverage gaps (unmapped concept, or < 3 Lexibank Tai-Kadai or Austronesian languages)	13
Unjustified PKDs (attestation_score = 1)	14
Unjustified PANs (attestation_score = 1)	27
Tier A (no coverage gap; PKD and PAN scores ≥ 2)	79

Categories are not disjoint: some pairs are both coverage-limited and score-1 on one side, and 6 pairs are unjustified on both the PKD and PAN sides. The important design point is that Tier A retains only slots where both reconstructions clear a minimal attestation bar against modern Lexibank evidence.

Among pairs that received attestation scores, PKD scores were distributed as 14 / 31 / 11 / 27 / 34 for scores 1 through 5 respectively ($n = 117$ scored). PAN scores on the stratified Austronesian sample were 27 / 27 / 12 / 14 / 31 ($n = 111$ scored). Thus a non-trivial fraction of the published package—especially on the Austronesian side under our sampling scheme—was judged poorly supported by daughters and was withheld from the chance test of form pairing.

Within Tier A, PKD attestation scores were predominantly 2–5 (25, 9, 22, and 23 pairs at scores 2–5), and PAN scores were likewise 2–5 (24, 11, 13, and 31). Tier A is therefore not restricted to only the most secure reconstructions; it excludes the weakest Layer 1 cases while still including many middling scores.

5.2 Observed meaning-blind hits

On the 79 Tier A pairs, the meaning-blind judge assigned generosity ≥ 4 to 27 pairs (34.2%). Of these hits, 18 scored 5 and 9 scored 4.

Illustrative high-scoring form pairs (generosity 5) include near-identity or near-identity modulo length/diacritics such as 1sg PAN *aku ~ PKD *aku; ‘die’ *(*m/p-*)aCay ~ *p-ata:y; ‘eat’ *ka(?)ən ~

*(i-)kan; ‘moon’ *bulaN ~ *bu[a:l; and ‘nine’ *siwa ~ *(s(i)wa:). These examples are listed for transparency; the statistical claim below concerns the **aggregate** hit count, not any single etymology.

5.3 Permutation null

Across **100** random reassignments of Tier A PAN forms to Tier A PKD slots, the number of hits had mean **5.68**, median **6**, and range **1–12** (first–third quartiles approximately 4–7). The observed count of **27** lies far above every null draw: no permutation produced 27 or more hits. The one-sided *p*-value is therefore

[*p* = .]

Under this screen, the published pairing yields roughly **four to five times** as many high form-similarity scores as the average null world (27 vs 5.7), and about **twice** as many as the most extreme null world in the sample (27 vs 12).

5.4 Sensitivity

Holding the empirical null distribution fixed, we asked how far the observed hit count would have to fall before *p* exceeded 0.05. With this null, an observed count of **10** still yields *p* ≈ 0.030, whereas **9** yields *p* ≈ 0.069. Relative to the actual **27** hits, that corresponds to discarding at least **18** of the 27 hits (two-thirds of them) before the result would cease to be significant at the 5% level under this null sample. This is a sensitivity calculation, not a new experiment: if entire slots were removed from Tier A, the null would also change. It does indicate, however, that the gap between observed and null is not a fragile one-hit margin.

5.5 Interim summary (Study 1)

After excluding coverage gaps and reconstructions scored as unsupported by Lexibank daughters, Smith’s (2025) remaining PAN–PKD alignments show meaning-blind form similarity well above a shuffled baseline (*p* ≈ 0.01, *N* = 100).

5.6 Study 2: Lexibank dual-attested concepts (Blust filter)

Concept set

Intersecting Lexibank concepts with ≥15 Tai-Kadai and ≥15 Austronesian languages (**373**) with the Blust/ABVD 210 Concepticon list yielded **194** analysis concepts. Coverage is high for core vocabulary (many concepts have 60+ Tai-Kadai and 500+ Austronesian languages before sampling).

Observed set-versus-set scores

On the **194** concepts, generosity scores were dominated by 1–2 (73 and 104 concepts respectively), with **6** scoring 3 and **11** scoring 4 (none scoring 5). Primary hits (generosity ≥ 4) are: *eye*, *nose*, *eat*, *water*, *bite*, *shoulder*, *mother*, *this*, *we*, *blow (of wind)*, and *be dead or die*. Adding generosity 3 brings in *fire*, *I*, *wing*, *ten*, *dust*, and *lightning* (**17** total at ≥ 3). At the liberal ≥ 2 cut, **121** concepts qualify—useful for sensitivity, but too inclusive for a primary claim.

Judge notes for ≥ 4 hits typically cite recurring skeletons familiar from classical comparisons (e.g. *mata*-like clusters for ‘eye’; *nam* / *danum*-like shapes for ‘water’; *kVn* for ‘eat’; *kat*-like shapes for ‘bite’). We treat these as transparency, not as etymological endorsements.

Permutation null

Across **30** random reassignments of Austronesian form groups to Tai-Kadai slots, hit counts were:

Threshold	Observed	Null mean	Null range	One-sided p
≥ 2	121	91.7	80–103	0.032
≥ 3	17	5.7	1–15	0.032
≥ 4	11	2.2	0–6	0.032
≥ 5	0	0.2	0–1	1.0

No null world reached the observed count at ≥ 2 , ≥ 3 , or ≥ 4 , so each of those p -values equals the add-one floor $1/31 \approx 0.032$. The ≥ 4 gap is especially clear: observed **11** versus null mean **2.2** (max **6**). At ≥ 2 the absolute null is high (~ 92), as expected for a liberal threshold, but the observed count still sits above every null draw.

Relation to Study 1

Study 2's ≥ 4 hit rate ($11/194 \approx 5.7\%$) is lower than Study 1's Tier A hit rate ($27/79 \approx 34\%$), which is expected: comparing diverse modern form groups is a harder screen than comparing two curated proto-strings. The important parallel is directional and statistical: in both designs, same-slot / same-meaning pairings exceed a form-preserving shuffle. Study 2 does so **without** relying on Smith's choice of reconstructions.

5.7 Summary of findings

Study 1 finds that Smith's remaining PAN–PKD alignments, after Lexibank attestation filters, show meaning-blind form similarity far above a shuffled baseline ($p \approx 0.01$). Study 2 finds that dual-attested Lexibank form inventories on a Blust concept filter likewise exceed a group-shuffle null at generosity ≥ 2 , ≥ 3 , and ≥ 4 ($p \approx 0.032$, $N = 30$). The next section discusses what this does and does not imply for the Austro-Tai hypothesis.

Figure: Null distribution

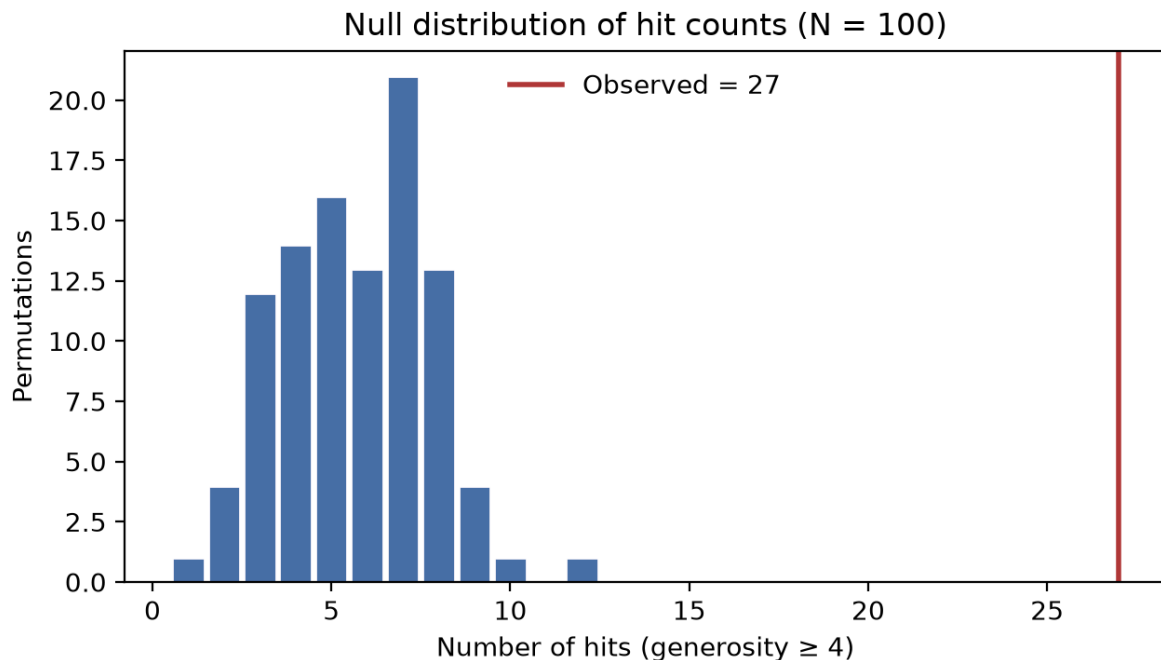


Figure 1: Null histogram of hit counts with observed count marked

Figure 2. Distribution of hit counts under 100 random reassignments of Tier A PAN forms to Tier A PKD slots. Red line: observed hits (27).

6. Discussion

6.1 What the results do and do not show

After Lexibank-facing filters, Smith's (2025) remaining PAN–PKD alignments show meaning-blind form similarity far above a shuffled baseline ($p \approx 0.01$ under $N = 100$ permutations). Independently, dual-attested Lexibank form groups on a Blust/ABVD concept filter also exceed a group-shuffle null at primary thresholds ($p \approx 0.032$ under $N = 30$). The natural reading is that **excess form resemblance under these screens is not an artifact of a single reconstruction spreadsheet**: Study 2 never sees Smith's proto-forms, yet still finds same-meaning form groups more similar than cross-meaning form groups.

That is a claim about **strength of evidence under stated controls**, not a demonstration of genetic relatedness. Classical arguments for Austro-Tai (especially Ostapirat 2005, 2013) turn on **regular phonological correspondences** and reconstructibility. Our judges may note plausible segment matches in free text, but they do not enforce a correspondence system, reconstruct intermediate stages, or distinguish inheritance from old loans that have been internalized in daughter vocabularies. Contact with Sinitic and mainland Southeast Asian languages remains a live alternative for some lookalikes. Excess aggregate hits therefore motivate further comparative work—including correspondence inventories over Study 2's hit concepts—they do not finish it.

Relative to the two main modern phylogenetic frames—Ostapirat's sister-family construal and Sagart's Austronesian-linked architecture—the results are **compatible with “there is non-chance lexical signal to explain”** and **agnostic** on tree geometry.

6.2 Why Study 2 is the more robust check

Study 1 answers: given Smith's published alignments, and after removing reconstructions Lexibank daughters do not support, is the residual pairing surprising? That is the right question for auditing a **fixed evidence package**, and anchoring in Smith reduces researcher freedom in picking etymologies (§2.2). It does not fully answer a different worry: that the package itself may preferentially include lookalike-friendly reconstructions.

Study 2 addresses that worry by construction. Concepts are selected by Lexibank coverage and a pre-existing Austronesian basic-vocabulary list (Blust/ABVD), not by an Austro-Tai author's alignments. Forms are modern Lexibank attestations under phylogenetic sampling, not proto-strings chosen for comparison. The null asks whether **same-Concepticon** inventories look more alike than shuffled inventories. Convergence between Study 1 and Study 2 therefore strengthens the claim that the signal is not merely an artifact of reconstructing toward resemblance.

The two studies are not interchangeable. Study 1 speaks directly to a published Austro-Tai spreadsheet; Study 2 speaks to modern lexical inventories and is closer to a “no reconstructions” stress test. Hit rates are lower in Study 2, as expected when comparing heterogeneous daughter samples rather than two curated proto-forms.

6.3 Why meaning-blindness and filters matter

If meanings are visible, both humans and models can reward “good etymologies” they already associate with famous comparisons, or stretch similarity when the gloss invites a match. Withholding glosses forces the score to be a function of the strings (or form groups) and the model's prior over orthographic/phonological resemblance. Caching binds observed and null worlds to the same scoring function.

In Study 1, Layer 1 addresses unsupported reconstructions. In Study 2, coverage floors, Blust filtering, onomatopoeia/dedupe heuristics, and phylogenetic sampling play the analogous role of

limiting opportunistic form choice—while still leaving room for within-family diversity and contact-induced lookalikes.

6.4 Limits

Several limits bound interpretation.

Model and prompt. All scores use gpt-4.1 via a fixed chat API and fixed prompts. Other models or stricter prompts would change absolute hit rates; the permutation comparison is meaningful only relative to the same scoring function. Threshold choice matters especially in Study 2: ≥ 2 yields many hits and a high null baseline; ≥ 4 is sparse but cleanly separated from the null.

Prior knowledge of etymologies. Large language models may have seen published Austro-Tai comparisons in training data. Meaning-blindness reduces semantic priming but does not erase memorized form pairs (e.g. famous *mata* / *nam* clusters). The permutation null still constrains the argument: memorization would have to favor **matched** pairings over shuffled ones. Residual risk remains and should be checked with expert human re-rating of hits.

Lexibank and sampling. Coverage is uneven across concepts and subgroups. Stratified samples avoid Oceanic overweighting but can miss reflexes. Study 2's onomatopoeia heuristics are imperfect. Blust intersection by Concepticon ID misses some near-matches where Lexibank splits Blust glosses.

Smith's preliminary character (Study 1). The Zenodo package is preliminary and versioned. Future releases may revise reconstructions; our frozen counts apply to v1.1 as parsed.

Null design and N. Study 1 uses $N = 100$; Study 2 uses $N = 30$ (costly set-versus-set calls). Study 2's $p \approx 0.032$ is therefore the add-one floor: no null reached observed, but a finer tail would need larger N . Neither null models phonetic natural classes, semantic fields, or borrowing pathways.

Nursery forms and deixis. Some Study 2 hits (*mother*, *this*, pronouns) are categories where chance or nursery resemblance is a known risk; primary interpretation should weight body-part and verb hits more heavily pending correspondence work.

6.5 Future work

Useful extensions include larger N for Study 2's null tail; human expert re-rating of hits; exploratory **regular correspondence** inventories over Study 2 hit concepts (at generosity ≥ 4 , ≥ 3 , and ≥ 2); correspondence-constrained scoring (encoding Ostapirat-style systems as hard filters); and parallel audits of other published Austro-Tai lists under Study 1's two-layer design.

7. Conclusion

We reported two meaning-blind permutation screens of Austro-Tai lexical resemblance. **Study 1** audited Alexander D. Smith's (2025) gloss-aligned PAN-PKD reconstructions with Lexibank attestation filters: on 79 Tier A pairs, 27 high form-similarity hits stand against a null mean of about 5.7 ($p \approx 0.01$, $N = 100$). **Study 2** compared dual-attested Lexibank Tai-Kadai and Austronesian form groups on 194 Blust/ABVD concepts, without reconstructions: hits at generosity ≥ 4 / ≥ 3 / ≥ 2 were 11 / 17 / 121 against null means of about 2.2 / 5.7 / 91.7 ($p \approx 0.032$, $N = 30$). Study 2 is the more robust check against cherry-picked reconstructions because concept and form selection do not inherit an author's comparative spreadsheet. Together, the studies strengthen the case that excess form resemblance under these screens is not merely an artifact of meaning-matched browsing—while leaving genetic proof to systematic sound correspondences and classical reconstruction.

Appendix

Companion files in the software release: output/judgments_observed.csv, data/unjustified_pairs.tsv, data/unjustified_pan_pairs.tsv, output/permutation_results.json, output/lexibank_attestation_audit.csv.

Appendix A. Prompt texts

A.1 Meaning-blind form similarity (prompt version v2)

System prompt (src/judge.py):

You are a comparative linguist scoring how similar proposed proto-language **forms** look, for exploratory Austro-Tai research.

Rules:

- Compare **phonetic and segmental shape only**. Do NOT use, infer, or mention meaning, semantics, glosses, or cognate labels.
- Each item gives a Proto-Kra-Dai (PKD) form and a candidate Proto-Austronesian (PAN) form. Score how similar the PAN form would be to the PKD form **if** it were the Austronesian reconstruction paired with that PKD slot—without knowing what concept either form represents.
- Do NOT require established regular sound correspondences; be generous on shape but not absurd.
- Ignore tone marks/superscripts when comparing shapes, but mention them if relevant.
- If the PAN side lists multiple alternate reconstructions, treat any plausible segmental match as supporting a higher score.

Return ONLY valid JSON (no markdown fences) as an array of objects with keys:

- comparison_id (string; echo the id from the input)
- generosity (integer 1-5; 5 = very similar shapes under a generous comparison)
- plausible_sound_correspondences (string; brief note or "none noted")
- reasoning (string; 1-3 sentences; do not refer to meaning)

User prompt pattern:

For each item, score how similar the proto-Austronesian form would be to the proto-Kra-Dai form if it were the Austronesian reconstruction for the same slot. Compare shapes only; do not use meaning.

+ JSON array of objects with keys: comparison_id, proto_austronesian, proto_kra_dai, proto_tai_branch (opaque batch index as comparison_id; no gloss).

A.2 PKD attestation vs Lexibank Tai-Kadai (prompt version v2)

System prompt (src/reconstruction_validate.py):

You are a comparative linguist assessing how well a proposed Proto-Kra-Dai (PKD) reconstruction is supported by attested modern Tai-Kadai word forms from Lexibank.

Rules:

- Compare the PKD form to attested daughter-language forms for the same meaning.
- Ignore tone marks/superscripts when comparing shapes, but mention them if relevant.
- Reward systematic reflexes across multiple languages; penalize if attested forms cluster around a different shape.
- Do NOT require perfect regular sound correspondences, but note major mismatches.
- If branch-level proto-Tai (PT) is given and contradicts most attested forms, lower the score.

Return ONLY valid JSON (no markdown fences) as an array of objects with keys:

- pair_id (string)
- attestation_score (integer 1-5; 5 = strongly supported across attested forms)
- supporting_reflexes (string; brief note on languages/forms that fit)
- problematic_reflexes (string; forms that do not fit, or "none noted")
- reasoning (string; 2-4 sentences)

User prompt pattern:

Score how well each PKD reconstruction is supported by the attested Tai-Kadai forms listed.

+ JSON array with pair_id, gloss, proto_kra_dai, proto_tai_branch, lexibank_concept, n_attested_languages_in_lexibank, attested_forms.

A.3 PAN attestation vs stratified Austronesian sample (prompt version v1)

System prompt (src/pan_validate.py):

You are a comparative linguist assessing how well a proposed Proto-Austronesian (PAN) reconstruction is supported by attested modern Austronesian word forms sampled from Lexibank.

Rules:

- Compare the PAN form to attested daughter-language forms for the same meaning.
- The sample is phylogenetically stratified (Formosan, Western Malayo-Polynesian, SHWNG, Oceanic, etc.). Weight evidence across clades; do not let Oceanic alone decide the score.
- Prefer support that includes Formosan and/or widespread Western MP reflexes for a high score.
- Ignore tone marks/superscripts when comparing shapes, but mention them if relevant.
- Reward systematic reflexes across multiple clades; penalize if attested forms cluster around a different shape.
- Do NOT require perfect regular sound correspondences, but note major mismatches.
- If alternate PAN reconstructions are listed, treat any well-supported alternate as supporting a higher score.

Return ONLY valid JSON (no markdown fences) as an array of objects with keys:

- pair_id (string)
- attestation_score (integer 1-5; 5 = strongly supported across sampled clades)
- supporting_reflexes (string; brief note on languages/forms/clades that fit)
- problematic_reflexes (string; forms that do not fit, or "none noted")
- reasoning (string; 2-4 sentences)

User prompt pattern:

Score how well each PAN reconstruction is supported by the sampled Austronesian forms listed.

Each attested form includes its coarse phylogenetic clade.

+ JSON array with pair_id, gloss, proto_austronesian, lexibank_concept_id, counts, attested_forms[{language,clade,form}].

Appendix B. Austronesian clade sampling

Language → clade map: data/lexibank/austronesian_language_clades.csv (coarse labels derived from Glottolog under Austronesian aust1307).

Sampling algorithm (src/an_sampling.sample_austronesian_forms):

1. Deduplicate to at most one form per language.

2. Satisfy per-clade floors when possible: Formosan **12**, Western Malayo-Polynesian **20**, SHWNG **8**, Oceanic **20**.
3. Fill remaining slots up to target **80** by round-robin across clades with leftover forms.
4. Use fixed seed AN_SAMPLE_SEED = 1 for reproducibility.

Appendix C. Tier A hits (generosity ≥ 4)

Observed hits: 27 / 79 Tier A pairs (18 with generosity 5; 9 with generosity 4). Null ($N = 100$): mean 5.68, range 1–12; one-sided $p = 0.0099$.

Gloss	PAN	PKD	Gen.	PKD att.	PAN att.
1sg	*aku	*aku	5	4	5
2sg	-Su; -mu (plural)	*-mə:	5	2	4
afraid; fear	*talaw	*C-la:w	5	4	4
dark; black	*-dəm	*(C_V /m)-dam	5	4	4
die	*(m/p-)aCay	*p-ata:y	5	4	5
drop; fall	*-tuq	*tok	5	5	2
eat	*ka(?)ən	*(i-)kan	5	4	5
fart	*qətut	*k[a/ə]tu2t	5	3	2
five	*lima	*[Vma:	5	3	5
hold; grab	*-[g/k]əm	*N-kam ^A	5	5	3
moon	*bulaN	*bu[a:l	5	2	5
nine	*siwa	*(s(i)wa:)	5	2	5
six	*(x)ənəm	*xən[a/ə]m	5	5	5
this	*-ni	*ni:	5	5	5
tongue2	*Səma	*Səma: ^A	5	4	2
tooth	*[l/n/ŋ]ipan	*(l)ipan	5	3	4
vomit	*utaq	*uta:k	5	2	4
water	*daNum	*danam	5	5	5
bird; chicken	*maNuk	*malok	4	4	5
deep	*daləm	*C-alək	4	5	2
excrement	<i>Caqi</i> ; (<i>PMP</i> taqay)	*CV-q[i/a:j]	4	2	3
fire	*Sapuy	*apuy	4	2	5
head	*quluh	*ku[ɭu:	4	2	5
louse	*kuCu	*C-uɭu:	4	2	5
shoulder	*qabaRa	*CV-ba:	4	5	5
sky	*Rabun	*C-bu1n	4	4	2
wash	*basəq	*C-sak	4	5	4

Appendix D. Unjustified reconstructions (attestation_score = 1)

D.1 Unjustified PKDs ($n = 14$)

Gloss	PAN	PKD
buy/sell	*saliw	*a j:w
leech	<i>matək</i> /məCaq	(*tak)
nest	*Rubu	*aru:k
otter	*Sanaq	*ana:k
pick up; lift; take	<i>saput</i> ; piliq	*kep
plant	*mula	*mu a:
sharp	*tazəm	*C.cəm ^A
shrimp	<i>qudaŋ</i> ; kabus	*(q)uda:ŋ
sick	*sakit	*ke:t
skinny	*Niwaŋ	*CV-ro:m
stomach	<i>biCuka</i> ; tiaN	*amok
tie; bundle	<i>Sikət</i> ; baluN; *bəjbəj	*CV-ru:k
to taste	*taɲam	*jim
turtle	*qaCipa (soft-shell)	*Ç-(i)pa: ^A

D.2 Unjustified PANs ($n = 27$)

Gloss	PAN	PKD
ant	<i>alujah</i> ; aNay	*Vmu2c
bathe	*diRus	*a:p
bear	*Cumay	*Cumaj
boat	*luja	*Çuda:
borrow	*Səzam	*Səzə:m
centipede	*qalu-Sipan	*CV-rip
child1	*aNak	*alə:k
dream	*S[ə/i]pi	*CV([+H])pan
fish poison	*tuba	*ÇV[+H]ba:
forget	*alim	*CV-lə:m
hook	<i>kawit</i> ; kabit	*C-bet
hot; warm	*lasuq	*C-ulu:l
hungry	(PMP *lapaR)	*C-ia:k
knife	<i>RabiS</i> ; tadaw	*mi:t
leg	*paqa	*paqa:
nest	*Rubu	*aru:k
plant	*mula	*mu[a:]
rattan	*quay	*(q)ua:y
raw; (a)live	*qudip	*(q)udip
shadow	<i>qaNiŋu</i> ; liŋaw	*aŋ[u/a:w]
sick	*sakit	*ke:t
skinny	*Niwaŋ	*CV-ro:m
sour	*qa(R)səm	*qas[a/ə]m
star	*qajaw	*ada:w
taro	*biRaŋ	*biRa:k
to taste	*taŋam	*jim
turtle	*qaCipa (soft-shell)	*Ç-(i)pa. ^A

Appendix E. Data and software availability

- **Smith (2025) reconstructions:** Zenodo DOI 10.5281/zenodo.15597357, release v1.1 (*Austro-Tai comparative dataset (Reconstructions)*).
- **Lexibank 2:** lexibank-analysed v2.2 (Concepticon-linked forms; Tai-Kadai and Austronesian subsets as described in §3–4).
- **Blust / ABVD concept list:** Concepticon contribution Blust-2008-210 (Greenhill, Blust & Gray 2008).
- **Analysis code and frozen outputs:** <https://github.com/jeremycollinsmpi/austro-tai-llm-test> (Study 1: parse → attest / validate-pan → judge → permute → report; Study 2: attested-core → attested-judge → attested-permute). Model: gpt-4.1 via project NLP chat endpoint. Frozen summaries: output/permutation_results.json (Study 1); output/attested_permutation_results_blust194_n30.json and output/attested_judgments_null_blust194_n30.csv (Study 2).

Appendix F. Study 2 hits (Blust dual-attested Lexibank)

Observed on **194** concepts; null $N = 30$; one-sided $p = 1/31 \approx 0.032$ at ≥ 2 , ≥ 3 , and ≥ 4 .

Generosity ≥ 4 ($n = 11$): eye; nose; eat; water; bite; shoulder; mother; this; we; blow (of wind); be dead or die.

Generosity $= 3$ ($n = 6$; total $\geq 3 = 17$): fire; I; wing; ten; dust; lightning.

Full scores and model notes: output/attested_judgments_observed_blust194.csv.

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